Hyeonjae Kim

Research Officer at the Agency for Defense Development

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Education

Kyungpook National University (KNU) — Daegu, Republic of Korea
Bachelor of Science in Physics — Mar. 2020 – Feb. 2024
Overall GPA: 4.11/4.3 (3.95/4.0) | Major GPA (71 credits): 4.19/4.3 (3.99/4.0)

Research Experiences

Agency for Defense Development (ADD) — Daejeon, Republic of Korea
Research Officer (Supervisors: Dr. Sin Hyuk Yim, Dr. Sangkyung Lee) — Jun. 2024 – May 2027 (expected)
Research in atomic physics and quantum sensing

- Designed and implemented an automated absorption spectroscopy system to characterize rubidium vapor cell properties
- Built and optimized a zero-field (SERF) optically pumped magnetometer for magnetic noise measurements
- Conducted magnetic-anomaly detection experiments and data analysis using a portable single-beam optically pumped magnetometer

Novel Applied Nano Optics Lab — Daegu, Republic of Korea
Research Intern (Advisor: Prof. Junyeob Yeo) — Jun. 2021 – Jun. 2022
Project: Fabrication of a flexible photo-electrochemical cell using laser

High Energy Physics Lab (Moon Lab) — Daegu, Republic of Korea
Research Intern (Advisor: Prof. Chang-Sung Moon) — Jun. 2020 – Feb. 2021
Developed momentum reconstruction algorithms for silicon detector simulations

Publications

UAV-based magnetic anomaly mapping of an undersea tunnel using a single-beam pulsed optically pumped magnetometer
Manuscript in preparation
H. Kim, S. Lee, and S. H. Yim

Novel sealing of glass-blown atomic gas cells using resistive heating wires
Manuscript under review
S. H. Hong, Y. Lim, S. H. Yim, S. Lee, **H. Kim**, T. Jeong, and J. B. Nam

Aging test of an atomic vapor cell with Al_2O_3 wall coating on cubic glass
Applied Optics **64**, 7932-7937 (2025)
H. Kim, T. Jeong, S. H. Hong, J. B. Nam, S. Lee, Y. Lim, and S. H. Yim

Selected Conference Presentations

Probing magnetic noise from portable sensor components using a zero-field optically pumped magnetometer
APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting, 2026 (scheduled)

Experimental systems for spectroscopy and zero-field OPM with atomic vapor cells
KISTI-SNU Joint workshop (QNRC Joint Workshop), 2026 (invited)

Towards the development of a SERF magnetometer
Korean Physical Society (KPS) Fall Meeting, 2025

Lifetime extension of rubidium vapor cells by Al_2O_3 coating
APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting, 2025

Linear absorption spectroscopy in rubidium vapor cells: Applications in buffer gas measurement and lifetime estimation
Korean Physical Society (KPS) Spring Meeting, 2025

Honors and Awards

Physics Alumni Association Award and Scholarship (Outstanding Graduate) Feb. 2024
Department of Physics Alumni Association, Kyungpook National University

2nd Prize, 2023 MiliTECH Challenge Dec. 2023
Hosted by KAIST; Awarded by the Agency for Defense Development

2nd Prize, 2022 MiliTECH Challenge Dec. 2022
Hosted by KAIST; Awarded by the Korea Military Academy (KMA)

3rd Prize of Academic Conference Dec. 2021
College of Natural Sciences, Kyungpook National University

Dean's List: 3 semesters (2021-1, 2021-2, 2022-2) Nov. 2021, Apr. 2022, Apr. 2023
College of Natural Sciences, Kyungpook National University

Grants and Scholarships

National Science and Technology Scholarship Mar. 2022 – Feb. 2024
Korea Student Aid Foundation (KOSAF)

Academic Merit Scholarship (Hyoseok Scholarship) Apr. 2022
Kyungpook National University Alumni Association

Academic Excellence Scholarship Aug. 2020, Feb. 2021, Aug. 2021
Kyungpook National University

Other Research Outputs

Buffer Gas Pressure Estimation for OPAMs Dec. 2024
Registered software; Korea Copyright Commission (C-2024-047156)

Rubidium Vapor Cell Lifetime Monitoring and Analysis Tool Dec. 2024
Registered software; Korea Copyright Commission (C-2024-054414)

Additional Experiences

Research Officer for National Defense (ROND) Jun. 2024 – May 2027
First Lieutenant in the Republic of Korea Air Force
Selected as one of 20 research officers nationwide for STEM-based national defense research

Teaching Assistant, Classical Mechanics II (PHYS212) Sep. 2023 – Dec. 2023

Teaching Assistant, Electromagnetism II (PHYS312) Mar. 2023 – Jun. 2023

International Student Tutor Feb. 2022 – May. 2022

Skills

Programming: Python, Wolfram Language (Mathematica)

Software: LabVIEW, SolidWorks, COMSOL Multiphysics